

Hamed Nilforoshan-Dardashti

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EDUCATION

Stanford University	Palo Alto, CA
Ph.D. Computer Science Advisor: Professor Jure Leskovec	2020 – 2026
Dissertation: “Real-world causal inference for healthcare and public policy”	
Columbia University	New York, NY
B.A. Computer Science GPA: 4.106, <i>Summa Cum Laude</i> (Top 5%)	2015 – 2019

HONORS

Best Paper Award (Outstanding Paper) - ICML 2022
Stanford Knight-Hennessy Scholarship - \$365,000 - Three years of grad. tuition, room, and board (< 2% acceptance rate)
Stanford Human-Centered AI Grant - \$100,000 - Grant supporting compute resources for AI for healthcare research
NSF Graduate Research Fellowship - \$138,000 - Three years of financial support for graduate research
Phi Beta Kappa, Columbia University - Awarded to top 10% of Columbia College graduates
Snapchat Research Scholarship - \$10,000 - Awarded to 8 students annually for outstanding coursework and research depth
Columbia John Jay Scholarship - Awarded to top 1% of Columbia College applicants
Columbia Computer Science Russell Mills Award - \$256 - Awarded to 4 students annually for academic excellence in CS
Columbia Venture Competition Finalist - Top 5 startups founded by Columbia students, out of over 100 companies
Columbia Fastpitch Winner - \$1,500 - Awarded 2nd place from over 60 companies pitched
Columbia NRHH (National Residence Hall Honorary) - Best Residential Community, for founding Comedy House
National Merit Scholar - \$2,500

EXPERIENCE

CEO and Co-Founder, Hiring.Cafe	2024 – 2026
Building a real-time dataset of every job on earth, and the AI to match every individual to the perfect job.	
Graduate Research Assistant, Stanford University	2020 – 2026
Data science and AI methods for healthcare, computational chemistry, and public policy [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11].	
Machine Learning Data Scientist, Airbnb Inc.	2019 – 2021
Developed and launched algorithms to mitigate theft, fraud, and money-laundering on the Airbnb marketplace.	
Research Intern, Snapchat Research	2018 – 2019
Designed and patented a graph model and algorithm to identify fraud rings (e.g. scams, fake news, botnets) on Snapchat [10].	
Research Assistant, Stanford University	2017 – 2019
Studied how environmental factors affect food consumption, and proposed interventions to reduce diet health inequalities [9].	
Data Science Intern, Airbnb Inc.	2018
Studied racial discrimination against African-American guests on Airbnb, and prototyped methods to reduce it.	
Research Assistant, Columbia University	2015 – 2018
Designed and evaluated novel AI methods to generate automated writing feedback [11, 12].	

Co-Founder and Project Lead, Bites App, Inc.	2015 – 2016
Developed a food marketplace app for iOS and Android, providing fresh, affordable, home-cooked meals to college students.	
Education Coordinator, Massachusetts Institute of Technology	2015 – 2015
Taught entrepreneurship to students at MIT Launch, a summer program and startup incubator for high-school students.	
Citizen Teacher, Citizen Schools	2013 – 2014
Taught programming and video game design to Boston public schools students at the Orchard Gardens Middle School.	
Founder and Lead Engineer, AlNoor Games	2012 – 2015
Built and published multiple web games with over 100,000,000 cumulative plays, while still in high-school.	

INVITED TALKS/PANELS

Massachusetts Institute of Technology	2024
Stanford University, Graduate School of Business Data Driven Seminar	2024
Stanford University, Causal Inference (CS 31N)	2023
Cornell University, Urban Data (INFO 5430)	2022
University College of London	2022
Technische Universität Darmstadt	2022
Encode Justice	2022
Airbnb, Inc.	2018
University of California, Irvine	2018
Buzzfeed, Inc.	2018

REVIEWING AND SERVICE

Reviewer, TMLR	2022
Reviewer, NeurIPS Workshop on Neuro Causal and Symbolic AI	2022
Reviewer, FaccT	2022
Reviewer, ICML	2022
Reviewer, The Web Conference WWW	2021

PUBLICATIONS

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- [1] **Hamed Nilforoshan**, Julia Reisler, Erfan Jahanparast, Michael Moor, Steven N. Goodman, Stefan Wager, Jure Leskovec. "Revised Estimates of the Types and Durations of Long Covid Symptoms Based on Claims Records from 245 million US Patients." Under Review in *PNAS* <https://www.medrxiv.org/content/10.64898/2026.02.17.26346448v1>
- [2] Eric Riesel*, Tsach Mackey*, Hamed Nilforoshan, Minkai Xu, Catherine Badding, Alison Altman, Jure Leskovec, Danna E. Freedman. "Crystal structure determination from powder diffraction patterns with generative machine learning." *Journal of the American Chemical Society* <https://pubs.acs.org/doi/10.1021/jacs.4c10244>
- [3] **Hamed Nilforoshan***, Wenli Looi*, Emma Pierson*, Blanca Villanueva, Nic Fishman, Yiling Chen, John Solar, Beth Redbird, David Grusky, Jure Leskovec. "Human Interaction Networks Reveal that Large Cities Facilitate Segregation." *Nature* <https://www.nature.com/articles/s41586-023-06757-3>
- [4] **Hamed Nilforoshan***, Michael Moor*, Yusuf Roohani, Yining Chen, Anja Šurina, Michihiro Yasunaga, Sara Oblak, Jure Leskovec. "Zero-shot causal learning." NeurIPS 2023 **Spotlight**. <https://arxiv.org/abs/2301.12292>

[5] Sam Corbett-Davies*, Johann Gaebler*, **Hamed Nilforoshan***, Ravi Shroff, Sharad Goel. “The Measure and Mismeasure of Fairness.” JMLR 2023 **Best Papers Track**. <https://arxiv.org/abs/1808.00023>

[6] Sajjad Fouladvand*, Frederico Gomes*, **Hamed Nilforoshan**, Jiaxuan You, Morteza Noshad, Rok Susic, Jure Leskovec, Jonathan Chen. Graph-Based Clinical Recommender: Predicting Specialists Procedure Orders using Graph Representation Learning. *Journal of Biomedical Informatics*. <http://hamedn.com/docs/clinical-recsys22.pdf>

[7] **Hamed Nilforoshan***, Johann Gaebler*, Ravi Shroff, Sharad Goel. “Causal Conceptions of Fairness and their Consequences.” ICML 2022 **Outstanding Paper Award** and Long talk [**< 0.2% acceptance rate**]. <http://hamedn.com/docs/causal-fairness22.pdf>

[8] Rishi Bonmassami,..., **Hamed Nilforoshan**,..., Percy Liang (116 authors). “On the Opportunities and Risks of Foundation Models.” Arxiv 2021. <https://arxiv.org/abs/2108.07258>

[9] Tim Althoff, **Hamed Nilforoshan**, Jenna Hua, Jure Leskovec. “Large-Scale Diet Tracking Data Reveal Disparate Impacts of Food Environment.” Nature Communications 2021. [**< 10% acceptance rate**] <http://hamedn.com/docs/diet21.pdf>

[10] **Hamed Nilforoshan**, Neil Shah. “SliceNDice: Mining Suspicious Multi-attribute Entity Groups with Multi-view Graphs.” in IEEE DSAA 2019. Washington, D.C. [**29% acceptance rate**] <http://hamedn.com/dsaa-19.pdf> + **2 patents pending**

[11] **Hamed Nilforoshan**, Eugene Wu. “Leveraging Quality Prediction Models for Automatic Writing Feedback.” in AAAI ICWSM 2018. Stanford, CA. [**16% acceptance rate**] <http://hamedn.com/dialectic-icwsm18.pdf>

[12] **Hamed Nilforoshan**, James Sands, Kevin Lin, Rahul Khanna, Eugene Wu. “Segment-Predict-Explain for Automatic Writing Feedback.” in Collective Intelligence 2017. New York, NY. <http://hamedn.com/docs/dialectic-ci17.pdf>

(* indicates co-first authorship)